



ASDE: Low-budget text classification via active semi-supervised learning with debiasing training mechanism

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ABSTRACT

Semi-supervised learning (SSL) is widely employed in text classification to address the challenges associated with limited labeled data availability. Nevertheless, current SSL methods often exhibit two significant limitations: they typically neglect the crucial process of selecting the initial labeled data effectively, and they fail to adequately mitigate the inherent bias that accumulates due to error propagation during the semi-supervised training phase. To address these shortcomings, we introduce an Active Semi-supervised learning framework with a DEbiasing training mechanism (ASDE). Specifically, ASDE includes a novel task-aware cold-start active data selection component designed to establish a more representative and informative initial labeled set by leveraging task-specific information. Additionally, to combat the detrimental effects of error propagation, we develop a spatial interpolation debiasing mechanism integrated into the training process. Empirical results on four widely used text classification datasets demonstrate the substantial performance gains achieved by our proposed ASDE framework, particularly under low-budget conditions.

1. Introduction

As a fundamental task in Natural Language Processing, text classification (Kowsari et al., 2019) serves as the backbone of modern information management systems. Its applications permeate critical domains including but not limited to: automated customer service systems requiring real-time topic categorization (Qu, Cong, Li, Sun, & Chen, 2012), social media monitoring platforms conducting sentiment analysis for brand reputation management (Wankhade, Rao, & Kulkarni, 2022; Zhou, Zhu, He, & Hu, 2016), and cybersecurity infrastructures implementing spam detection (Rao, Verma, & Bhatia, 2021). Text classification enables intelligent organization of scientific repositories (Cunha et al., 2020), precision retrieval of legal documents (Shi, Hu, Xie, Guo, & Wu, 2024), and dynamic filtering of misinformation in digital libraries (Abburi et al., 2023). The deployment of deep learning architectures has indeed advanced these applications through complex pattern recognition (Li et al., 2023), yet their supervised learning paradigms demand increasingly prohibitive quantities of annotated data (Sun, Qiu, Xu, & Huang, 2019).

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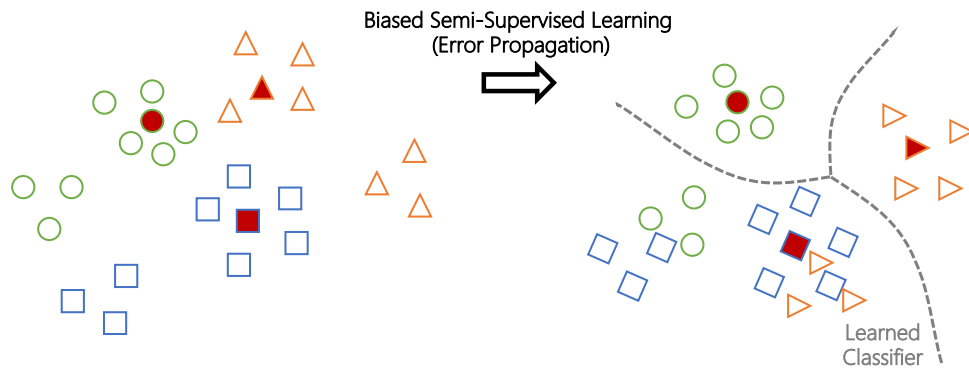


Fig. 1. The illustration of the manifold collapsing in low-budget semi-supervised learning by error propagation. The data points with the same shape stand for the same gold category. The left part shows the ideal representation space. When the most representative data points colored in red are selected, an SSL method tends to annotate unlabeled data points. Once the data points shaped in Circles or triangles are wrongly categorized into squares, the model will intensify these since these errors are hard to be perceived.

However, the real-world implementation of these systems often confronts severe data scarcity challenges. In medical information processing, for instance, labeling clinical notes requires domain expertise and faces strict privacy constraints, making large-scale annotation prohibitively expensive and slow (Chen et al., 2025; Meystre, Savova, Kipper-Schuler, & Hurdle, 2008). Financial institutions developing fraud detection models must handle proprietary jargon that resists crowdsourced annotation. Efficient learning with minimal labeled examples is crucial for timely adaptation to new fraud patterns (Wei, Li, Cao, Ou, & Chen, 2013). Even in consumer applications, multilingual content analysis suffers from inadequate low-resource language corpora (Nekoto et al., 2020). These practical scenarios expose the critical bottleneck of existing methods: their dependence on large labeled datasets conflicts with the economic and technical realities of specialized domains (Halevy, Norvig, & Pereira, 2009). The annotation dilemma intensifies for tasks demanding nuanced interpretation, such as distinguishing sarcasm in sentiment analysis (Joshi, Bhattacharyya, & Carman, 2017) or identifying emerging misinformation patterns (Zubiaga, Aker, Bontcheva, Liakata, & Procter, 2018), where even expert annotations may exhibit significant subjectivity (Krippendorff, 2018). This confluence of high annotation costs and task complexity frequently leads to a scarcity of labeled data, thereby hindering the effective training and deployment of supervised systems. These real-world constraints across diverse domains highlight the critical need for advanced text classification methods that can perform effectively with minimal labeled data. Addressing the problem of text classification under low-resource or limited-labeling conditions is thus of paramount importance.

Semi-supervised learning (SSL) has emerged as a critical paradigm to mitigate the reliance on extensive labeled datasets, a significant bottleneck in numerous machine learning applications (Berthelot et al., 2019; Sohn et al., 2020; Tarvainen & Valpola, 2017). SSL effectively leverages readily available unlabeled data by transferring knowledge acquired from a limited set of labeled examples. This knowledge transfer is commonly facilitated through several key techniques. Consistency-based regularization, for instance, promotes robust models by encouraging consistent predictions for perturbed versions of unlabeled inputs (Miyato, Maeda, Koyama, & Ishii, 2018; Sajjadi, Javanmardi, & Tasdizen, 2016; Tarvainen & Valpola, 2017). Pseudo-labeling assigns artificial labels to unlabeled data based on high-confidence model predictions, subsequently incorporating these into the training process (Lee et al., 2013; Mukherjee & Awadallah, 2020; Zou, Yu, Liu, Kumar, & Wang, 2019). Entropy minimization aims to enhance the discriminative power of the model on unlabeled data by encouraging low-entropy, high-confidence predictions (Berthelot et al., 2019; Grandvalet & Bengio, 2004). By synergistically combining supervised signals from labeled data with unsupervised information derived from the inherent structure of unlabeled data, SSL provides a powerful framework for learning under limited supervision.

Despite its promise, the practical application of SSL is confronted by two primary challenges. The first is the critical issue of Initial Labeled Data Selection: the composition and representativeness of the initial small labeled subset profoundly influence the performance of SSL algorithms. A poorly chosen subset may not adequately capture the underlying data distribution, potentially leading to biased or suboptimal model learning. The second challenge is Error Propagation: during the training process, particularly when employing techniques like pseudo-labeling, initial prediction errors on unlabeled data can be amplified and disseminated throughout the dataset. This can result in the model becoming overconfident in incorrect predictions, ultimately degrading overall performance. This issue is particularly exacerbated when the initial labeled dataset is small or contains noise.

Data Selection: The remarkable efficacy of state-of-the-art semi-supervised learning methods, achieving performance levels comparable to fully supervised approaches when trained on extensive datasets, critically hinges on the availability of a sufficiently large and diverse initial set of labeled data instances (Chen, Yang, & Yang, 2020; Chen, Zhou, Li & Zhao, 2025; Xie, Dai, Hovy, Luong, & Le, 2020). However, the effectiveness of SSL deteriorates significantly when the labeling budget is severely constrained. Under such low-resource conditions, the performance becomes highly sensitive to the specific composition of this initial labeled subset, exhibiting substantial variability and potentially suboptimal outcomes across different random selections. This vulnerability arises because the initial labeled data serves as the fundamental basis for model initialization, pseudo-labeling, and subsequent learning iterations. Even a small, strategically chosen initial labeled set can disproportionately influence the learning trajectory and

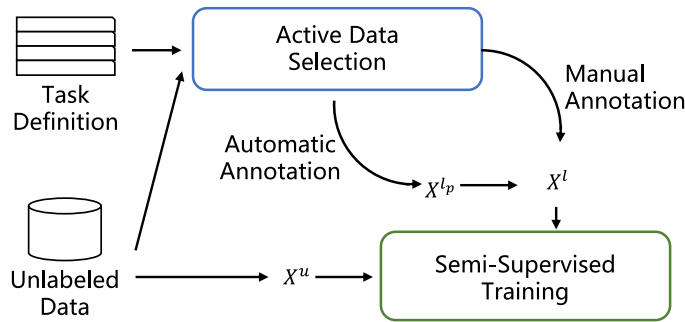


Fig. 2. The overall framework of our Active Semi-Supervised Learning.

the achievable performance ceiling. Consequently, identifying an optimal initial labeled set is paramount for enabling effective SSL when annotation resources are scarce. Active learning (AL) presents a promising paradigm to mitigate this challenge by intelligently selecting data points for annotation, aiming to maximize model performance within a limited labeling budget (Chakraborty & Singh, 2022; Cohn, Atlas, & Ladner, 1994; Sener & Savarese, 2018; Wei, Iyer, & Bilmes, 2015; Yoo & Kweon, 2019). AL strategies are broadly categorized into model-dependent and model-independent approaches (Seo, Kim, Ahn, & Lee, 2022; Yuan, Lin, & Boyd-Graber, 2020). Model-dependent AL methods iteratively query instances considered most informative or uncertain by the current model. While effective in scenarios with a moderate amount of initial labeled data, these methods inherently face a “cold-start” problem when the initial budget is extremely low, as training a reliable initial model is difficult. This dependency on a pre-trained model limits their applicability and stability in truly low-resource settings. Conversely, model-independent AL approaches aim to select representative data points based on the underlying data distribution without relying on an initial task-specific model. These often leverage unsupervised or self-supervised techniques to learn data representations (e.g., using pre-trained language models (Devlin, Chang, Lee, & Toutanova, 2019)) and then employ strategies like clustering to identify diverse or central instances. A significant limitation here is the reliance on representations that may be task-agnostic. Representations learned through general pretext tasks or from large pre-trained models might not fully capture the task-specific nuances that determine an instance’s informativeness or its true label distribution within a cluster. Therefore, the assumption that selecting cluster centroids or diverse points based on these representations guarantees selecting instances informative for the target task or accurately representing label space is not always valid (Chapelle, Weston, & Schölkopf, 2002).

A core challenge in low-budget SSL, particularly addressing the cold-start problem, lies in effectively selecting an initial labeled set that simultaneously balances *informativeness* (crucial for rapid model improvement) and *representativeness* (essential for covering the data manifold and capturing the overall distribution). Existing AL paradigms struggle to optimally achieve this balance from the outset when the available initial labeled data is minimal, necessitating novel approaches for initial data selection under extreme budget constraints.

Error Propagation: Predicted label distributions are a crucial component in semi-supervised learning, enabling knowledge transfer from a limited labeled dataset to a large unlabeled dataset. This mechanism underpins various SSL paradigms, such as pseudo-label construction via thresholding or voting in self-training (Zou et al., 2019), distribution sharpening in consistency regularization (Chen et al., 2020), and entropy estimation for boundary regularization (Berthelot et al., 2019). However, error propagation remains a significant challenge in SSL. Initial misclassifications on unlabeled data can be amplified throughout training, leading to overconfident yet inaccurate subsequent predictions (Lokhande, Tasneeyapant, Venkatesh, Ravi, & Singh, 2020). This creates a detrimental feedback loop where the model reinforces its own mistakes, resulting in a substantial discrepancy between predicted confidence and actual model accuracy. This error propagation issue is particularly exacerbated in low-budget scenarios where the number of labeled examples is severely restricted. As illustrated conceptually in Fig. 1, even strategic selection of representative data points may not provide comprehensive coverage of the underlying data distribution or its sub-clusters. Consequently, deep learning models, which prioritize minimizing loss on the available data, can become overly reliant on easily discriminable features present within the limited labeled set. This leads to erroneous generalization, where the model predicts incorrectly on unlabeled instances that share these superficial features but belong to different true classes. We posit that this phenomenon contributes to the formation of a biased or degenerate representation space. This space reflects a skewed understanding of the data, characterized by a non-uniform or collapsed distribution of categories, driven by the model’s focus on the sparse and potentially unrepresentative labeled data. This representational bias is further compounded by error propagation, as amplified misclassifications reinforce these distorted feature representations. Therefore, effectively mitigating error propagation and debiasing the learning process are critical challenges in the context of low-budget SSL.

To mitigate the aforementioned challenges, we present a novel Active Semi-Supervised Learning framework incorporating a Debiasing Training mechanism. This approach aims to enhance the robustness and generalizability of the learned model, particularly in scenarios with limited labeled data. We introduce a task-aware cold-start active data selection strategy designed to optimize the initial data labeling process. This strategy is particularly crucial for low-resource semi-supervised text classification as it ensures the selection of a highly representative initial labeled dataset, thereby boosting model performance when labeled data is scarce.

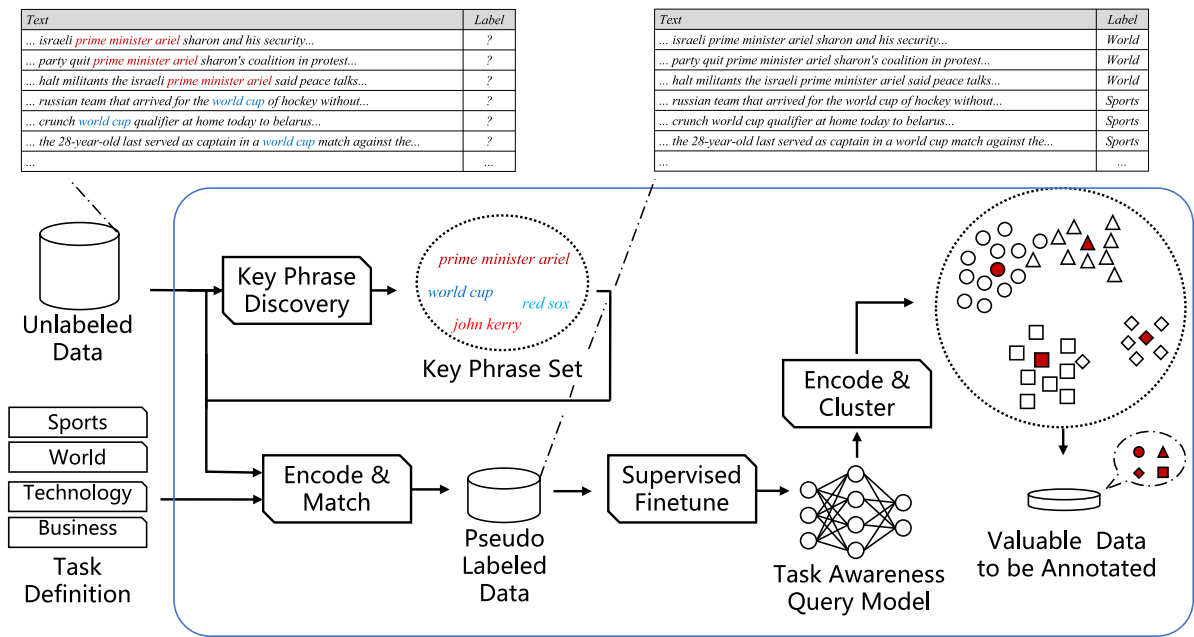


Fig. 3. The workflow of task awareness cold-start data selection module.

This strategy leverages task-specific knowledge to guide the selection of a maximally informative and representative initial labeled dataset. We balance two key criteria: informativeness, assessed through task-relevant signals, and representativeness, evaluated by the density of data points in the representation space. This dual consideration ensures the selected data are both highly relevant to the target task and effectively capture the underlying data distribution. Specifically, we employ a pre-trained language model to identify high-frequency task-aligned key phrases within the unlabeled corpus. These key phrases facilitate the automatic labeling of a subset of representative texts, which then train a task-aware query model. Unlabeled data are subsequently clustered based on their hidden representations and predicted class probabilities. To manage practical labeling constraints, the uncertainty threshold for active selection is dynamically adjusted according to the available budget. Furthermore, to mitigate error propagation, particularly under limited labeling budgets in semi-supervised text classification, we propose a spatial interpolation-based debiasing training mechanism. This mechanism aims to maximize the representational separation between unlabeled data points predicted to belong to the same class. By encouraging a more uniform and dispersed distribution of unlabeled data within the representation space, this approach effectively alleviates the manifold collapse phenomenon often observed when training with limited and potentially biased labeled data. The contributions of this paper are as follows:

- We propose a novel active semi-supervised learning framework seamlessly integrating a debiasing training mechanism to simultaneously tackle data selection and error propagation challenges in low-budget text classification.
- We introduce a task-aware cold-start active data selection method that effectively identifies the most informative and representative data points for annotation, specifically designed to address the data selection challenge.
- We devise a spatial interpolation-based debiasing training mechanism that guides the model to learn a more unified and less collapsed representation manifold, thereby tackling the error propagation issue.
- Extensive experiments on four widely used text classification datasets demonstrate that our proposed method significantly outperforms state-of-the-art approaches, particularly in low-budget settings.

2. Related work

The pursuit of efficient and effective text classification, particularly in low-resource scenarios, has driven substantial research across several interconnected domains. This section reviews pertinent advancements in text classification methodologies, semi-supervised learning (SSL), and active learning (AL), highlighting the context and motivation for our proposed active semi-supervised debiasing approach, ASDE.

2.1. Text classification

Text classification, the task of assigning predefined categorical labels to textual units (Liu, Wang, Li, & Liu, 2010; Xia, Yin, Feng, & Philip, 2021; Zhou, Huang, Hu, & He, 2020), has been profoundly impacted by the advent of deep learning. Early

deep learning approaches, such as Convolutional Neural Networks (CNNs) adept at capturing local n-gram features (Chen, 2015), and Recurrent Neural Networks (RNNs) designed to model long-range sequential dependencies (Tai, Socher, & Manning, 2015), demonstrated considerable improvements over traditional machine learning and rule-based systems. The paradigm shifted further with the emergence of large-scale pre-trained language models (PLMs) like BERT and ERNIE (Ding et al., 2021; Sun et al., 2019). Fine-tuning these PLMs on downstream tasks has become the de facto standard, achieving state-of-the-art performance across a multitude of text classification benchmarks.

Recent efforts continue to push the boundaries of PLM-based text classification. For instance, Balkus and Yan (2024) explored the use of generative models like GPT-3 to synthesize training data, thereby augmenting performance for short text classification. Concurrently, Pan, Hang, Sil, and Potdar (2022) introduced a contrastive adversarial training method to enhance the robustness and generalization of fine-tuned Transformer encoders by learning noise-invariant representations. Despite these advancements, the predominant supervised learning paradigm heavily relies on substantial quantities of labeled data. This dependency poses a significant bottleneck in many real-world applications where data annotation is expensive or time-consuming, creating a critical need for methods that can perform effectively under data scarcity, a challenge our work directly addresses.

2.2. Semi-supervised learning

Semi-supervised learning (SSL) offers a compelling avenue to mitigate the label scarcity problem by incorporating vast amounts of readily available unlabeled data alongside limited labeled instances to enhance model performance (Jiao et al., 2024; Li et al., 2019; Yang, Song, King, & Xu, 2022). SSL has garnered particular attention in text classification due to the natural abundance of unlabeled text (Chen et al., 2023, 2020; Mukherjee & Awadallah, 2020; Xu, Chen, Huang, Vucetic, & Yin, 2024; Xu, Liu, & Abbasnejad, 2022; Yang, Tu, Lu, Yin, & Chow, 2015).

Two prominent SSL strategies are self-training and consistency regularization. Self-training methods (Arazo, Ortego, Albert, O'Connor, & McGuinness, 2020; Chi, Tang, Lu, Zhang, & Jang, 2023; Xie, Luong, Hovy & Le, 2020) typically involve training an initial model on labeled data, using it to generate pseudo-labels for unlabeled data, and then retraining the model on a combination of original labels and high-confidence pseudo-labels, often iteratively. While effective, self-training can be susceptible to confirmation bias and error propagation if pseudo-labels are noisy or biased. Consistency-based regularization methods (Lee, Hahn, Seo, Park, & Han, 2024; Li, Liu, & Tan, 2019; Mukherjee & Awadallah, 2020), on the other hand, leverage data augmentation techniques (Wei & Zou, 2019). These methods train models to produce consistent predictions for an unlabeled instance and its augmented versions, thereby encouraging the learning of robust, invariant features. The efficacy of these methods often hinges on the quality and diversity of augmentations and the underlying assumption that perturbations maintain semantic integrity.

Some SSL approaches attempt to incorporate domain or task-specific knowledge. For example, Xu et al. (2022) integrated knowledge from PLMs and class names into the SSL training process. However, their method incorporates task definition information primarily during the SSL training phase rather than upfront in the data selection or active learning component. Our work, ASDE, explores an alternative strategy by infusing task information directly into the data selection process within an active semi-supervised framework, aiming for more targeted and efficient learning, particularly in low-budget settings, and incorporating mechanisms to counteract potential biases introduced during SSL.

2.3. Active learning

Active learning (AL) aims to maximize model performance while minimizing annotation costs by strategically selecting the most informative unlabeled instances for labeling (He, Jin, Ding, Yi, & Yan, 2019; Li et al., 2024; Wang & Shang, 2014). This is particularly crucial in low-budget scenarios, including the cold-start setting where no initial labeled seed samples are available.

AL strategies are broadly categorized into uncertainty-based and diversity-based (or distribution-based) approaches. Uncertainty-based methods query instances where the current model exhibits the highest prediction uncertainty. While intuitive, these methods often require a reasonably well-calibrated model trained on initial labeled seed samples, which may not be feasible in strict low-budget or cold-start scenarios. Furthermore, they can be prone to selecting outliers or focusing on noisy boundary regions.

Distribution-based or representativeness-based methods (Sener & Savarese, 2018; Seo et al., 2022; Yuan et al., 2020) seek to select a small yet representative subset of the entire unlabeled data pool, often by clustering data representations and selecting points near cluster centroids. The quality of these selected instances heavily depends on the learned representations. If these representations, typically derived from self-supervised pretext tasks, are not well-aligned with the target task's underlying data manifold, the selected samples may not be optimally informative or could even introduce biases.

Recent research has explored combining AL with SSL. For instance, Wang, Lian, and Yu (2021) proposed a method that integrates AL data selection with SSL for image classification by clustering representations from self-supervised learning. Our approach, ASDE, builds upon this direction by specifically tailoring the synergy between AL and SSL for text classification. We advance this by incorporating task-specific information more deeply into the active data selection phase, even in cold-start settings. This aims to overcome the limitations of generic representations and uncertainty measures, thereby selecting more relevant and potentially less biased data. The subsequent debiasing training mechanism in ASDE is designed to further refine the learning process, addressing biases that may arise from the small initial set of actively selected data or during the semi-supervised propagation of information. This integrated approach is pivotal for achieving robust and reliable text classification with minimal labeled data.

In summary, existing approaches either (i) rely on a pre-trained model for active selection, which fails in cold-start settings; (ii) select data based on generic representations, ignoring task-specific semantics; or (iii) lack a mechanism to mitigate error propagation during SSL training. ASDE addresses these limitations by introducing a task-aware cold-start strategy and a spatial debiasing mechanism, distinguishing itself from prior work.

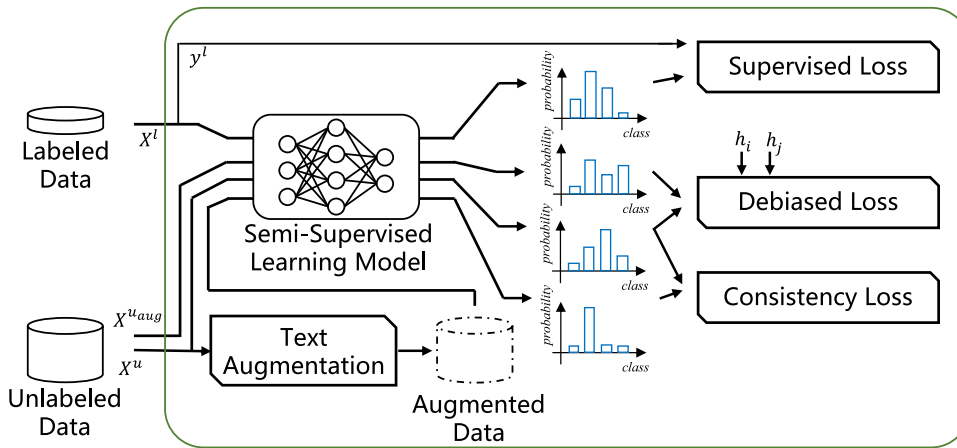


Fig. 4. The workflow of semi-supervised learning with spatial interpolation debiasing training mechanism.

3. Method

Building upon the challenges outlined in Section 1, this section introduces our proposed framework (ASDE), which integrates active learning with semi-supervised learning under a unified debiasing mechanism (see Fig. 2). The following sections detail the two core components: task-aware cold-start data selection (Section 3.1) (see Fig. 3) and spatial interpolation-based debiasing training (Section 3.2) (see Fig. 4). Fig. 2 presents the overall architecture of ASDE. A central aspect is the active selection of initial data, which is used for both human labeling and generating automatic pseudo-labels, forming the basis for semi-supervised training.

Given a text classification dataset comprising an unlabeled data pool $X = \{X_1, \dots, X_n\}$ and a defined set of target classes $C = \{C_1, \dots, C_m\}$, our data selection module D initiates the process. Utilizing information derived from the task definition C , D actively selects a small, informative, and representative subset of data points, denoted as $X^l = \{X_1^l, \dots, X_{n_l}^l\}$, for subsequent human annotation, yielding ground truth labels y^l . Concurrently, as a component of the selection process, D identifies and assigns reliable pseudo-labels y^p to another subset of data points sampled from the unlabeled pool, $X^p = \{X_1^p, \dots, X_{n_p}^p\}$. We then combine the human-labeled data X^l and pseudo-labeled data X^p to form the composite training set $X' = [X^l; X^p]$. This combined set is subsequently fed into the semi-supervised learning module S . The training of S incorporates three key objectives: a supervised loss on the human-labeled data X^l , consistency regularization leveraging the larger unlabeled data pool, and our proposed spatial interpolation debiasing mechanism. The primary objective of this training regimen is to effectively transfer knowledge from the limited labeled data to the extensive unlabeled data while simultaneously mitigating the propagation of errors, particularly those arising from pseudo-labeling.

3.1. Task awareness cold-start active data selection

Pre-trained Language Models (PLMs) have demonstrated remarkable capabilities by learning extensive linguistic knowledge from vast text corpora through self-supervised objectives (Azaria, Azoulay, & Reches, 2024; Chen, 2024; Chen, Zhang, et al., 2025; Zhao, Zhou, Jin, Yuan, Chen, Liu, & Li, 2024). However, a key challenge is that without fine-tuning on a specific downstream task, the representations learned by PLMs tend to capture general linguistic features rather than those specifically relevant to the task at hand (Chen, Zhou, et al., 2025). This can limit their performance when applied directly to specialized applications. To address this, we aim to explicitly incorporate task definitions into the PLM's framework. By doing so, we seek to better leverage the model's inherent capacity to encode features that are specifically tailored to the requirements of the target task. This approach is expected to yield more effective and task-oriented representations, thereby improving performance on downstream NLP applications.

Since a general PLM can hardly find task-relevant parts of a plain text to focus on, some prior statistical information could be mined and introduced to PLM. We mine the task/dataset specific textual segments (key phrases).

To identify important terms and phrases within the text corpus, we employ a two-pronged approach focusing on both unigrams and multi-word expressions. For selecting significant unigram keywords, we utilize the well-established Term Frequency-Inverse Document Frequency (TF-IDF) weighting scheme (Aizawa, 2003). TF-IDF assigns a score to each term based on its frequency within a specific document relative to its frequency across the entire corpus. This balance helps to filter out common words while highlighting terms that are particularly relevant to a given document. The standard TF-IDF calculation is as follows:

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \quad (1)$$

$$idf(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|} \quad (2)$$

$$tfidf(t, d, D) = tf(t, d) \cdot idf(t, D) \quad (3)$$

where $f_{t,d}$ is the raw count of term t in document d . $\sum_{t' \in d} f_{t',d}$ is the total number of terms in document d . N is the total number of documents in the corpus D . $|\{d \in D : t \in d\}|$ is the number of documents in D that contain term t .

Terms with high TF-IDF scores are selected as candidate unigram key phrases, indicating their potential significance within a document and rarity across the corpus.

Beyond individual terms, characteristic phrases often carry domain-specific meaning, such as “*prime minister ariel*” in a political context. To identify these salient multi-word expressions, we incorporate a method based on the statistical association between adjacent words. Pointwise Mutual Information (PMI) is employed to measure this association, quantifying how much more likely two words co-occur than if they were independent. High PMI scores suggest a strong statistical dependency between adjacent words, indicating they likely form a meaningful phrase or collocation.

The PMI between two words x and y is calculated as:

$$PMI(x; y) = \log \frac{p(x, y)}{p(x)p(y)} \quad (4)$$

where $p(x, y)$ is the joint probability of words x and y occurring together (typically as adjacent words). $p(x)$ and $p(y)$ are the individual probabilities of words x and y occurring in the corpus, respectively.

Word combinations exhibiting high PMI values are considered strong candidates for multi-word phrases, reflecting their internal coherence and potential significance within the dataset. These identified phrases, along with the selected unigram keywords, contribute to a richer representation of the textual content.

The algorithm calculates the left and right information entropy as the degree of freedom of the extraction results from PMI to filter the redundancy substrings further. Extraction results with high entropy will be reserved as n -gram key phrases.

$$E_{left}(PreW) = - \sum_{\forall Pre \subseteq A} P(PreW) \log P(PreW) \quad (5)$$

After mining the key phrase set, we construct a mask M for every data according to the presence of these key phrases, key-phrase-tokens are set to 1, and others are 0. When getting the text representation from PLM, we average vectors encoded by PLM where the value of the corresponding mask is 1.

$$h^{key}(X_i) = AvgPool(PLM(X_i) \cdot M) \quad (6)$$

The task definition of text classification is the class names in plain text. They can be further complemented by definition in Wikipedia. We get the match score s of text and all candidate class C descriptions by PLM, to further filter the data X_i with s_{ij} exceeding the threshold value τ labeled as class j .

$$s(X_i, C_j) = cosine(h^{key}(X_i), PLM(C_j)) \quad (7)$$

Hacohen, Dekel, and Weinshall (2022) indicates that different difficulty samples' contributions would change with the budget size. Easy samples are more important in low-budget settings, and on the contrary, hard samples may be more valuable when the budget increases. We emphasize a low-budget setting and expect to introduce task information without any manual label information for initial data selection. So these data X^{lp} with relatively simple and explicit textual features to match the target class would benefit our query model even for the semi-supervised training process. A query model PLM' is finetuned by X^{lp} on the foundation of PLM. Then, the mixture representation of text can be obtained by concrete the task specific representation $h_i = PLM'(X_i)$, and the prediction probability distribution p_i .

$$h_i^{cluster} = [h_i; p_i] \quad (8)$$

We apply K-Means++ (Arthur & Vassilvitskii, 2006) on these vectors and set K as the budget size. The centroid in each cluster is selected as informative and representative data X^l to be annotated.

3.2. Spatial interpolation debiasing semi-supervised training mechanism

We introduce our debiasing training mechanism on the basis of UDA (Mukherjee & Awadallah, 2020). As shown in Fig. 4, we optimize the SSL model with the consideration of three directions: supervised loss, debiased loss, and consistency loss.

Supervised Learning: To enable the model to effectively learn the primary task of text classification, we incorporate a supervised loss function. Specifically, for our binary classification problem, we utilize the Binary Cross-Entropy (BCE) loss. This standard loss function quantifies the discrepancy between the model's predicted probability for the positive class and the corresponding ground truth binary label. By providing a direct gradient signal derived from the labeled data, the supervised loss is crucial for guiding the model's initial parameter updates and steering it towards learning discriminative features relevant to the classification task, especially in the early stages of training when other learning signals might be weak. The supervised BCE loss, calculated as the average over a batch of n_l labeled instances, is defined as:

$$\mathcal{L}_l = \frac{1}{n_l} \sum_{i=1}^{n_l} [-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)] \quad (9)$$

Table 1

Statistics of four text datasets.

Dataset	Label type	Classes	Unlabeled	Dev	Test
AG News	News Topic	4	20,000	2,000	7,600
DBpedia	Wikipedia Topic	14	70,000	7,000	70,000
Yahoo!	QA Topic	10	50,000	6,000	60,000
IMDB	Review Sentiment	2	10,000	2,500	25,000

where $y_i \in \{0, 1\}$ represents the ground truth binary label for the i th instance, and $\hat{y}_i \in [0, 1]$ is the model's predicted probability for the positive class.

This signal is activated exclusively when the prediction confidence falls below a dynamic threshold η_t , which is determined at training step t . For supervising this optimization process, both manually annotated and actively selected data X^l and automatically labeled pseudo-data X^{lp} are utilized as labeled examples.

Consistency Learning: In semi-supervised learning, the consistency loss leverages unlabeled data by encouraging the model to produce similar output distributions for an input and its perturbed (augmented) version. This objective promotes robustness and helps transfer the knowledge encoded in the model parameters, which learned from labeled data, to the unlabeled data distribution. Specifically, for an unlabeled sample x_i^u , we minimize the discrepancy between the model's predicted probability distribution for the original sample, $p(y|x_i^u)$, and that for its augmented counterpart, $p(y|x_i^{u, aug})$. We employ the Kullback–Leibler (KL) divergence to quantify this discrepancy:

$$\mathcal{L}_u = \frac{1}{n_u} \sum_{i=1}^{n_u} KL(p(y|x_i^u), p(y|x_i^{u, aug})) \quad (10)$$

Here, n_u is the number of unlabeled samples in the batch, X_i^u denotes the i th unlabeled input, $X_i^{u, aug}$ represents a stochastically augmented version of X_i^u , and $p(y|x)$ is the model's predicted probability distribution over the output labels y given input X . Minimizing this loss encourages the model's predictions to be invariant to semantically-preserving transformations of the input, thereby improving generalization on unlabeled data.

Debiased Representation Learning: Existing SSL approaches, particularly those relying on consistency regularization, do not explicitly account for the potential propagation of errors during training on unlabeled data. As conceptually illustrated in Fig. 1, when the model makes an incorrect prediction on an unlabeled instance based on features learned from the limited labeled set X^l , the standard consistency loss may inadvertently reinforce this erroneous prediction unless data augmentations happen to resolve the ambiguity. As training progresses, such mistakes can be solidified and propagated to other unlabeled data points.

Furthermore, as depicted in Fig. 1, over-parameterized models trained with limited supervision can easily learn a class-symmetrical representation space that is locally effective for the labeled data but fails to ensure a well-structured or uniform distribution of unlabeled data globally. This can lead to ambiguity, misclassification, or overconfident wrong predictions for unlabeled examples situated in suboptimal regions of the feature space.

To address this challenge, one could increase the labeled data budget or employ more sophisticated data selection strategies. Alternatively, a complementary approach is to encourage a more structured and less biased distribution of unlabeled data in the representation space, simultaneously mitigating overconfidence in erroneous predictions. We pursue the latter strategy and propose a novel debiased Loss $\mathcal{L}_d$, acting as a form of spatial regularization in the representation space.

Specifically, the Debiased Loss is formulated as:

$$\mathcal{L}_d = \sum_{X_i^u \in X^u, X_j^u \in X^u, \text{s.t. } l(X_i^u) = l(X_j^u)} (1 - \cos(h(X_i^u), h(X_j^u))) \quad (11)$$

where $h(\cdot)$ denotes the feature embedding function, $l(\cdot)$ is the model's prediction function returning the predicted class label, and X^u is the set of unlabeled data. This objective minimizes the angular distance (equivalently, maximizes the cosine similarity) between the feature embeddings of pairs of unlabeled data points that are currently predicted to belong to the same class. By promoting compactness and similarity within the predicted clusters, this loss imposes a structural constraint on the feature space. This encouraged local structure helps to solidify correct cluster assignments and can implicitly penalize instances that are outliers within their predicted cluster, thereby reducing the model's confidence in potentially incorrect predictions and mitigating the reinforcing effect of error propagation.

Final Loss: The training object $\mathcal{L}$ is the combination of the above three losses.

$$\mathcal{L} = \eta_t \cdot (\mathcal{L}_l + \lambda_d \cdot \mathcal{L}_d) + \lambda_u \cdot \mathcal{L}_u \quad (12)$$

where $\eta_t = \frac{t}{T} * (1 - \frac{1}{K}) + \frac{1}{k}$ refer to the linear schedules of TSA in Mukherjee and Awadallah (2020).

Table 2

Comparison of our model and other baselines on for text classification datasets in different labeled data per class. Each experiment is conducted over 5 independent trials, and each time we use different random seeds for initialization. We report the mean $\pm$ standard deviation of each result.

Dataset	Method	Number of Labeled Sample Per Class			
		3	5	10	20
AG News	BERT-FT	76.70 $\pm$ 4.72	79.90 $\pm$ 2.34	83.46 $\pm$ 2.73	84.97 $\pm$ 1.73
	UDA	78.25 $\pm$ 7.61	82.97 $\pm$ 2.87	86.75 $\pm$ 0.88	86.77 $\pm$ 0.10
	MixText	81.60 $\pm$ 9.04	85.84 $\pm$ 1.32	85.56 $\pm$ 2.95	87.60 $\pm$ 0.48
	PCM	84.85 $\pm$ 0.86	87.20 $\pm$ 0.42	88.31 $\pm$ 0.47	88.34 $\pm$ 0.27
	SuperST	84.93 $\pm$ 0.45	87.5 $\pm$ 0.57	87.74 $\pm$ 0.54	88.49 $\pm$ 0.24
	ASDE(Ours)	85.62 $\pm$ 2.20	87.70 $\pm$ 0.98	87.38 $\pm$ 0.65	88.61 $\pm$ 0.19
DBpedia	BERT-FT	86.68 $\pm$ 2.59	91.86 $\pm$ 2.46	96.60 $\pm$ 0.46	97.84 $\pm$ 0.23
	UDA	93.51 $\pm$ 2.23	95.88 $\pm$ 2.78	97.26 $\pm$ 1.50	98.59 $\pm$ 0.04
	MixText	93.25 $\pm$ 0.68	96.93 $\pm$ 0.41	98.39 $\pm$ 0.09	98.64 $\pm$ 0.18
	PCM	94.37 $\pm$ 0.49	97.04 $\pm$ 0.68	98.70 $\pm$ 0.04	98.80 $\pm$ 0.06
	SuperST	94.45 $\pm$ 0.72	97.23 $\pm$ 0.37	98.84 $\pm$ 0.76	98.94 $\pm$ 0.35
	ASDE(Ours)	94.52 $\pm$ 0.61	97.55 $\pm$ 0.64	99.08 $\pm$ 0.18	99.02 $\pm$ 0.16
Yahoo!	BERT-FT	45.93 $\pm$ 3.67	50.75 $\pm$ 4.32	61.84 $\pm$ 2.37	63.89 $\pm$ 0.94
	UDA	48.30 $\pm$ 11.09	57.09 $\pm$ 5.69	65.15 $\pm$ 1.54	67.76 $\pm$ 0.60
	MixText	60.27 $\pm$ 4.29	65.77 $\pm$ 1.78	67.23 $\pm$ 1.97	68.19 $\pm$ 1.33
	PCM	63.52 $\pm$ 2.63	67.09 $\pm$ 0.54	68.34 $\pm$ 1.03	69.21 $\pm$ 0.42
	SuperST	64.23 $\pm$ 1.23	67.56 $\pm$ 0.56	68.38 $\pm$ 0.87	69.75 $\pm$ 0.47
	ASDE(Ours)	65.66 $\pm$ 2.64	68.21 $\pm$ 1.38	68.49 $\pm$ 0.63	70.07 $\pm$ 0.48
IMDB	BERT-FT	60.11 $\pm$ 2.41	65.17 $\pm$ 8.39	73.20 $\pm$ 2.97	78.70 $\pm$ 6.75
	UDA	63.01 $\pm$ 1.07	71.90 $\pm$ 10.80	89.05 $\pm$ 1.70	90.20 $\pm$ 0.54
	MixText	56.27 $\pm$ 3.46	71.89 $\pm$ 4.89	83.38 $\pm$ 3.35	86.27 $\pm$ 1.36
	PCM	73.86 $\pm$ 1.04	86.06 $\pm$ 0.74	89.94 $\pm$ 0.44	91.10 $\pm$ 0.28
	SuperST	74.35 $\pm$ 2.23	86.75 $\pm$ 1.82	88.94 $\pm$ 1.29	90.87 $\pm$ 1.14
	ASDE(Ours)	79.15 $\pm$ 2.09	88.50 $\pm$ 1.20	90.21 $\pm$ 0.18	91.03 $\pm$ 0.22

4. Experiments

4.1. Datasets and preprocessing

For fair comparisons, we follow the datasets used in previous works on semi-supervised text classification (Chen et al., 2020; Xu et al., 2022), including AG News (Zhang, Zhao, & LeCun, 2015), DBpedia (Lehmann et al., 2015), Yahoo! Answers (Chang, Ratino, Roth, & Srikumar, 2008), and IMDB (Maas et al., 2011). The dataset preprocessing process refers to Xu et al. (2022). We randomly select a subset as an unlabeled set for each dataset. The dataset statistics and split information are shown in Table 1.

We apply multiple types of augmentations for every unlabeled text in advance, including back-translation, unimportant word replacement by TF-IDF measurement, and synonym word replacement by Wordnet and PPDB. The intermediate languages of back-translation are France, German, Spanish, Chinese, Russian, and Arabic. We use Google Translation API to get translations. We randomly chose one type of augmented text during consistency training. We truncate all the input text into 256 to take into account both training efficiency and PLM's length limit. Note that for IMDB, we start to truncate from the last token rather than the first token as Mukherjee and Awadallah (2020).

4.2. Metrics and parameter settings

Since all test sets in these four datasets are class balanced, we evaluate our method using accuracy. PyTorch is chosen for our method's implementation (Paszke et al., 2019).

Previous work in SSL performed random sampling from the whole dataset to constitute the initial labeled set. The initially labeled sets are class balanced, with a pre-set number of samples per class. In practice, without awareness of the label of every data, to sample a balanced initial set with n samples per class in a dataset with m classes, the expectation of total sample count is larger than $m * n$. For a fair comparison to previous works on SSL, we set the budget size of active sampling b , in which the probability of failures (minimal class less than n) is less than 5%. We get b according to the program simulation. With b labeled data, we randomly select a balanced subset to conform to n samples per class for experiments.

This expectation and budget setting only take effect under the assumption that the dataset is class balanced. The data selection strategy and experiment settings on an unbalanced dataset with a low budget are left for our future work.

For hyperparameters, we perform a grid search to reach the trade-off between validation performance and training efficiency. We set the learning rate to $2e-5$ for all datasets. The additional classifier on the top of BERT is a two-layer MLP with a 768 hidden size and Tanh as an activation function. λ_d is 0.01, λ_u is 1. And we select the top 8 samples per class for pseudo label generation in the active data selection module. We select subsets of 5000 samples in all unlabeled data pools as the candidate for data selection.

Table 3

Ablation results on the IMDB test set.

Method	Number of Labeled Sample Per Class		
	3	5	10
Ours	79.15 $\pm$ 2.09	88.50 $\pm$ 1.20	90.21 $\pm$ 0.18
ours (modify data selection to random)	70.98 $\pm$ 5.50	83.83 $\pm$ 2.69	88.64 $\pm$ 0.46
ours (modify data selection to cls-cluster)	73.65 $\pm$ 2.71	86.93 $\pm$ 1.16	87.36 $\pm$ 0.16
ours (w/o spatial interpolation)	78.59 $\pm$ 1.67	87.11 $\pm$ 0.28	89.24 $\pm$ 0.14

4.3. Baselines

We compare our method with the following baselines:

BERT-FT: Directly finetune pre-trained BERT model with labeled data.

UDA (Mukherjee & Awadallah, 2020): Performing consistency training after data augmentations for semi-supervised training.

MixText (Chen et al., 2020): A semi-supervised method is designed to weighted mix two different inputs with their hidden representation at a random layer in the encoder, and the weighted label as supervision as X. It also uses data augment for consistency training.

PCM (Xu et al., 2022): Utilizing a k-way classifier and a matching classifier to introduce class semantic information in semi-supervised learning.

SuperST (Lee et al., 2024): utilizing “superficial learning” to low-confidence pseudo-labels and conventional fine-tuning to high-confidence pseudo-labels plus gold labels to integrate semi-supervised learning with large language models.

4.4. Comparative analysis with existing methods

The results of our model compared to other baselines are shown in Table 2. Across all datasets and methods, a clear trend emerges: performance generally improves as the number of labeled samples per class increases. This is an expected outcome, as more labeled data typically allows models to learn better representations and decision boundaries. However, the degree of improvement varies across methods and datasets, highlighting their respective strengths and weaknesses in low-resource versus relatively higher-resource settings.

As a fine-tuned BERT model without specific semi-supervised techniques, BERT-FT serves as a supervised baseline. Its performance is consistently the lowest across all datasets and data regimes, particularly in the few-shot settings (3 and 5 labeled samples). This underscores the challenge of training large models like BERT effectively with very limited labeled data and motivates the need for semi-supervised learning approaches. UDA shows significant improvements over BERT-FT, particularly in low-resource settings. This demonstrates the effectiveness of leveraging unlabeled data through techniques like data augmentation and consistency training. However, UDA’s performance gains seem to saturate or even slightly decrease at higher labeled data points on some datasets (e.g., AG News 10 vs 20 samples), and its standard deviations can be relatively high in the 3-shot setting (e.g., Yahoo! and IMDB), indicating instability. MixText, which utilizes MixMatch and Mixup for text, also shows strong performance, generally outperforming UDA and BERT-FT. It is particularly competitive in some low-resource scenarios. Its performance tends to be more stable than UDA in some cases (e.g., lower standard deviations on Yahoo! and IMDB 3-shot). PCM demonstrates strong results, often achieving the best performance among the baselines, especially in the 10 and 20 labeled sample settings. This suggests that incorporating contrastive learning with prototypical representations is effective for semi-supervised text classification. PCM shows good stability with generally low standard deviations across different data regimes.

The proposed ASDE method demonstrates highly competitive and superior performance compared to all baselines across the evaluated datasets and labeled data settings. In low-resource settings (3 and 5 labeled samples per class), ASDE consistently achieves the highest mean accuracy in the 3 and 5 labeled sample settings across all four datasets. This is a significant finding, indicating ASDE’s strong ability to learn effectively from very limited labeled data. The improvements over the next best baselines are notable, particularly on datasets like Yahoo! and IMDB where the few-shot problem is more challenging. In higher-resource settings (10 and 20 labeled samples per class), ASDE maintains its leading position in most cases even as the amount of labeled data increases. On DBpedia, ASDE achieves the best performance across all data regimes. On Yahoo! and IMDB, it consistently leads or is statistically tied for the best performance (considering the standard deviations). On AG News, ASDE is the best in the 3, 5, and 20-sample settings, while PCM is slightly better at 10 samples, though the difference is within the standard deviation.

The results strongly support the effectiveness of the proposed ASDE model for semi-supervised text classification, particularly in low-resource scenarios. ASDE’s consistent outperformance in the 3 and 5 labeled sample settings across diverse datasets is a key highlight, demonstrating its ability to leverage unlabeled data effectively when labeled data is scarce. This is crucial for many real-world applications where obtaining large quantities of labeled text data is expensive or difficult.

While ASDE maintains strong performance in settings with more labeled data, the gains over the best baselines like PCM become smaller, suggesting that as more labeled data becomes available, the relative advantage of different semi-supervised techniques may diminish as all models benefit from the increased supervision.

Table 4

Selected samples from our active data selection module.

Label	Text
Pos	dudley moore is fantastic in this largely unknown classic . . . worth seeing many times.
	. . . and i don't care if they're marketable. this movie always leaves me in tears. it's perfect . the end could not be better. and then watch the original trilogy yet again. bravo!
Neg	this is one of the worst films i have *ever* seen! it is bad , . . . the plot is diabolically flawed , and the known names in this film are wasted on confused, uncertain characters. . .
	. . . there's way too much story for such a stupid movie. like i said. i wouldn't even steal this movie. and for the person who says this is ' funniest movie of 2007'.....need to take a look around, the news is better than this.

4.5. Ablation experiments

To investigate the effectiveness of our proposed components of the method, we also perform the ablation experiments on the IMDB dataset. The ablation results are shown in Table 3, indicating that none of these models can achieve a comparable result with our full version. Demonstrate that all those factors contribute a certain improvement to our model. (1) Effectiveness of Task awareness cold-start active data selection. Specifically, when we discard the whole data selection part (random data selection), the performance drops demonstrate the effectiveness of the data selection. To further investigate the contribution of the task information in data selection, we replace our task awareness selection module with cls-cluster (cluster data points only based on the CLS token's representation in the PLM and select the centroid to label). The performance of this alternative is also inferior to our method, even not showing the superiority of the application of data selection. We find that the general representations encoded by PLM without task information lack discrimination for different classes. This further proves the necessity of introducing task information in data selection of low-budget text classification.

(2) Effectiveness of Spatial interpolation debiasing training mechanism When we remove the spatial interpolation operation, the accuracy slightly decreases, also indicating the effectiveness of our debiasing training mechanism. Note that the performance gain of our training mechanism is bigger in the SSL method without data selection. Since more potential bias would cause without carefully selecting the initial labeled data, the debiasing training mechanism would benefit more from potential error propagation in the SSL training process.

4.6. Discussion

We apply various visualization to explore our proposed active semi-supervised learning framework further. The discrimination of our key phrase context representation Fig. 5 shows the T-SNE (Van der Maaten & Hinton, 2008) visualization of the general representation of PLM and our proposed key phrase contextual representation of the unlabeled dataset. Different shape stands for different gold class. The density of data points in two-dimensional space also reflects the density of representation space to a certain extent. It manifests that our key phrase representation method is more faithful to the real label and can bring the task information in before the data selection.

(1) **The Selected Data Points.** Table 4 shows part of the data selected in our active data selection process. Those samples contain many label-related phrases like “fantastic”, “perfect” in positive class, “flawed”, and “stupid” for negative class. Typical samples like those are suitable as centroids to the corresponding class. (2) **Training Dynamics in SSL.** Fig. 5 is the transformation of representation space in an unlabeled dataset during SSL training, from the preliminary stage to the later period. The shape stands for the gold label, and the color represents the predicted label. Points with in star are initially labeled. The right part (Fig. 5b and d) omits the spatial interpolation debiasing training mechanism.

We find that the data points are more dispersed with our spatial interpolation debiasing training mechanism. The initial labeled data points (in a and b marked as star) are scattered all over the final representation space, manifesting the representativeness and informativeness of selected data. However, in d, the data representations are not closely clustered around the initial annotation points and the class centers. This indicates that the debias mechanism can guide the model's class representations to be more faithful to the actual label semantics during the training process.

Fig. 5 illustrates the evolution of the representation space during semi-supervised learning training. Specifically, Figs. 5a and 5c depict the representation space at a preliminary and a later training stage, respectively, when our proposed spatial interpolation debiasing mechanism is employed. In contrast, Figs. 5b and 5d show the representation space at the corresponding preliminary and later stages without the debiasing mechanism. In all panels, the shape of each point represents its gold label, while the color indicates the model's predicted label. Points marked with a star denote the initially labeled data points.

Comparing the panels, we observe distinct differences in the structure and distribution of the representation space attributable to the debiasing mechanism. In the later training stage (Figs. 5c vs. 5d), the data points are notably more dispersed when our spatial interpolation debiasing mechanism is active (Fig. 5c). Furthermore, the initially labeled data points (marked with stars), which serve as anchors, appear scattered throughout the final representation space in Fig. 5c, suggesting their diverse representativeness and informativeness in guiding the learning process across different regions of the data distribution. Conversely, in the absence of the debiasing mechanism (Fig. 5d), the data representations in the later stage appear less dispersed, and critically, the clustering of points around the initial annotation points and the natural class centers is less pronounced and coherent compared to Fig. 5c.

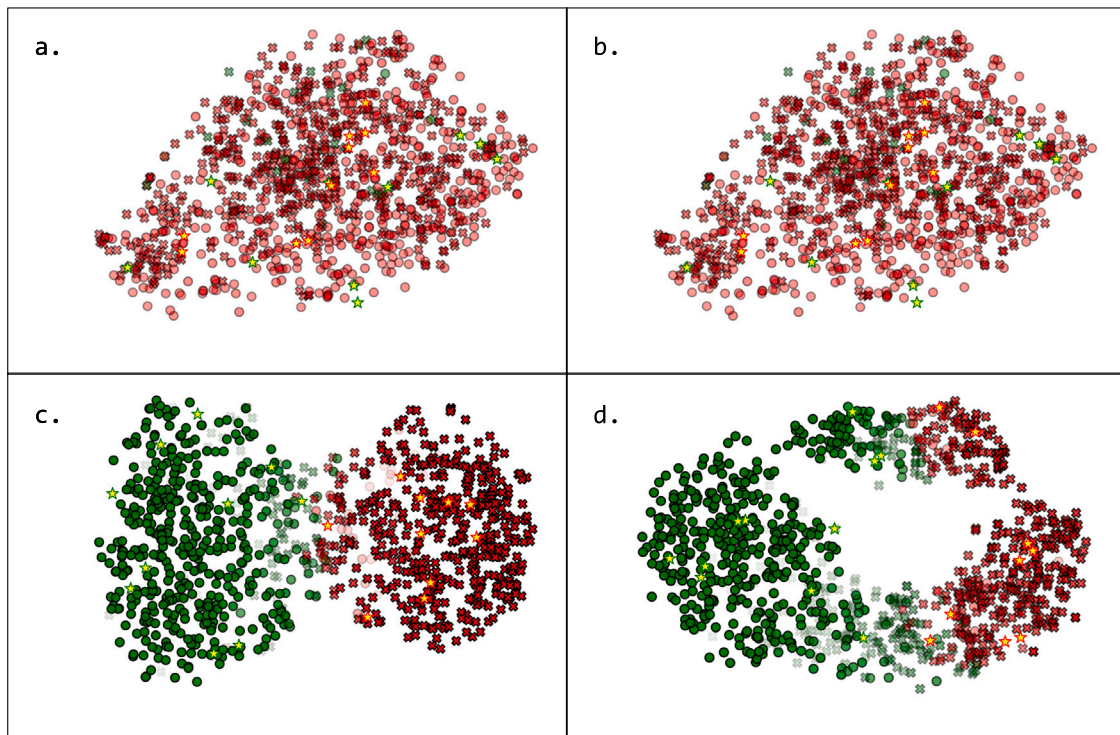


Fig. 5. Visualization of the representation space before and after SSL training.

This detailed comparison across the training trajectory highlights that the spatial interpolation debiasing mechanism effectively encourages the model to form class representations that are more spread out yet semantically consistent and better aligned with the underlying true label semantics, as evidenced by the improved clustering behavior relative to both initial labeled points and implicit class centers in the debiased setting.

5. Conclusion

In this paper, we propose an active semi-supervised learning framework designed for text classification tasks under limited label budgets. Our framework addresses key challenges: data selection for the initial labeled set and error propagation during semi-supervised training. To tackle the former, we introduced a task-aware active cold-start data selection module that effectively identifies both informative and representative data points for labeling. For the latter, our novel spatial interpolation debiasing loss was developed to mitigate error propagation by encouraging a more uniform and robust representation space. Comprehensive experiments conducted on four text classification datasets demonstrate that our proposed framework achieves significant performance improvements over existing state-of-the-art methods. By improving text classification accuracy under stringent labeling budgets, ASDE has the potential to lower the barrier for deploying NLP solutions in resource-constrained scenarios across healthcare, finance, multilingual content analysis, and beyond, where abundant unlabeled data exists but labeled examples are scarce or costly to obtain.

6. Limitations

A key limitation of semi-supervised learning methods is their susceptibility to parameter selection and the inherent stochasticity of the training process. Effective parameter tuning, including the selection of appropriate data augmentation techniques, conventionally relies on access to a large labeled development set. This dependence on labeled data represents a practical constraint, particularly in resource-constrained scenarios. The problem of dynamically balancing data allocation for training and evaluation under a unified budget therefore warrants further investigation.

CRedit authorship contribution statement

Yubo Chen: Writing – review & editing, Software, Project administration, Methodology, Funding acquisition, Conceptualization. **Tong Zhou:** Writing – original draft, Validation, Software, Methodology, Conceptualization. **Daojian Zeng:** Writing – review & editing, Validation, Software, Methodology. **Sirui Li:** Writing – review & editing, Software, Resources, Investigation. **Kang Liu:** Writing – review & editing, Supervision. **Jun Zhao:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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Data availability

Data will be made available on request.

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